Root locus procedure

1. Write the characteristic equation so that the variable parameter k appears as a multiplier
$$1 + kP(s) = 0.$$
2. Factor $P(s)$ in terms of n_p poles and n_z zeros: $1 + k\frac{\prod_{i=1}^{n_z}(s+z_i)}{\prod_{j=1}^{n_p}(s+p_j)} = 0$
(the zeros are: $-z_i$, $i = \overline{1, n_z}$ and the poles: $-p_j$, $j = \overline{1, n_p}$)
3. Locate the open-loop poles and zeros of $P(s)$ in the s-plane with symbols: $\times$ - open-loop poles, o - open-loop zeros.
4. Determine the number of separate loci SL. $SL = n_p$, when $n_p \geq n_z$, where n_p is the number of open-loop poles and n_z the number of open-loop zeros.
5. Locate the segments of the real axis that are root loci:
   (a) On the real axis the locus lies to the left of an odd number of open-loop poles and zeros
   (b) Locus begins at an open-loop pole and ends at an open-loop zero (or infinity along to asymptotes- in case the number of zeros is smaller than the number of poles)
6. The root locus is symmetrical with respect to the horizontal real axis.
7. The loci proceed to infinity along asymptotes centered at σ_A and with angles Φ_A.
$$\sigma_A = \frac{\sum(poles) - \sum(zeros)}{n_p - n_z} = \frac{\sum_{j=1}^{n_p}(-p_j) - \sum_{i=1}^{n_z}(-z_i)}{n_p - n_z}$$
$$\Phi_A = \frac{2q+1}{n_p - n_z} \cdot 180^o, \quad q = 0, 1, 2, ...(n_p - n_z - 1)$$
8. By utilizing the Routh-Hurwitz criterion, determine the point at which the locus crosses the imaginary axis (if it does so).
9. Determine the breakaway or break-in points on the real axis (if any)
   (a) Set $k = -\frac{1}{P(s)} = p(s)$, (from the characteristic equation $1 + kP(s) = 0$)
   (b) Obtain $dp(s)/ds = 0$
   (c) Determine roots of (b) or use graphical method to find maximum of $p(s)$.

If required:

1. Determine the angle of locus departure from complex poles and the angle of locus arrival at complex zeros, using the phase criterion
$$\angle P(s) = \pm 180^o(2q+1), \ at\ s = -p_j\ or\ -z_i.$$
2. Determine the root locations that satisfy the phase criterion
$$\angle P(s) = \pm 180^o(2q+1) \text{ at a root location } s_x$$
3. Determine the parameter value k_x at a specific root s_x using the magnitude condition:
$$k_x = \frac{\prod_{j=1}^{n_p} \mid s + p_j \mid}{\prod_{i=1}^{n_z} \mid s + z_i \mid} \Big|_{s=s_x}$$